10TH CLASS PHYSICS — NEW SYLLABUS (PECTAA)

Chapter 17 — Electromagnetism

Solved Exercise | New Book (PECTAA 2026)

Exercise 17 (Solved)

This exercise covers the magnetic effect of a current, the right-hand grip rule, the force on a current-carrying conductor in a magnetic field, the d.c. motor and the Earth's magnetic field.

A. Multiple Choice Questions

17.1 Which statement is true about the magnetic poles?

(a) Unlike poles repel
(b) Like poles attract
(c) Magnetic poles do not affect each other
(d) A single magnetic pole does not exist

Answer: (d) A single magnetic pole does not exist — like poles repel, unlike poles attract, and every magnet has both a north and a south pole — an isolated magnetic monopole has never been found.

17.2 A D.C motor converts:

(a) mechanical energy into electrical energy
(b) mechanical energy into chemical energy
(c) electrical energy into mechanical energy
(d) electrical energy into chemical energy

Answer: (c) electrical energy into mechanical energy — the current in the coil experiences a magnetic force which turns the coil and drives the load, so electrical energy is converted into mechanical (rotational) energy.

17.3 The part of a D.C motor that reverses the direction of current through the coil every half-cycle:

(a) the armature
(b) the commutator
(c) the brushes
(d) the springs

Answer: (b) the commutator — the split-ring commutator changes the connections of the coil to the brushes twice in every rotation, so the coil keeps turning in the same direction.

17.4 The two main layers of the Earth's core that contribute to the generation of the magnetic field are:

(a) crust and mantle
(b) inner core and outer core
(c) lithosphere and asthenosphere
(d) troposphere and stratosphere

Answer: (b) inner core and outer core — the motion of molten iron and nickel in the outer core around the solid inner core acts as a huge dynamo and produces the Earth's magnetic field.

17.5 What does a steady electric current in a conductor generate around it?

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(a) Electric field
(b) Light
(c) Magnetic field
(d) Sound waves

Answer: (c) Magnetic field — Oersted's discovery: a current always produces a magnetic field in the space around the conductor, and its direction is given by the right-hand grip rule.

17.6 What happens to the magnetic field direction when the direction of the current is reversed?

(a) The magnetic field remains unchanged
(b) The magnetic field also reverses direction
(c) The magnetic field becomes stronger
(d) The magnetic field disappears

Answer: (b) The magnetic field also reverses direction — the direction of the field depends on the direction of the current, so reversing the current reverses the field (the magnitude stays the same).

B. Short Answer Questions

17.1 What is the direction of the magnetic field around a straight current carrying wire?

The magnetic field around a straight current-carrying wire consists of concentric circles lying in a plane perpendicular to the wire, with the wire at the centre. The direction of these circular field lines is given by the right-hand grip rule: if the wire is held in the right hand with the thumb pointing along the current, then the curled fingers point in the direction of the magnetic field.

If the current flows upwards, the field lines are anticlockwise when seen from above; if the current is reversed, the field also reverses and becomes clockwise. The field is strongest near the wire and becomes weaker as we move away from it, since $B = \mu_0 I / (2\pi r)$.

17.2 What happens to the magnetic field if the magnitude of the current in a wire is increased?

The strength (magnitude) of the magnetic field increases. The field at a distance r from a long straight wire is $B = \mu_0 I / (2\pi r)$, so B is directly proportional to the current I. Doubling the current doubles the field.

The field lines become closer together near the wire, showing a stronger field, but their pattern and direction stay the same. If the current is reduced, the field becomes weaker; if it is switched off, the field disappears. In a solenoid (coil), increasing the current similarly increases the strength of the field inside the coil in the same proportion.

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17.3 How can you identify the north and south poles of an unmarked magnetized steel bar?

Suspend the bar horizontally from its middle by a fine thread and let it come to rest. A magnet is free to rotate, so it will set itself along the north–south direction of the Earth's field. The end that points towards the geographic north is the north pole of the magnet and the other end is the south pole.

This can be checked in other ways. Bring one end of the bar near the marked north pole of a known magnet: if it is repelled, that end is also a north pole; if it is attracted, it is a south pole. (Attraction alone is not reliable, because any pole attracts an unmagnetised piece of iron.) Another method is to use a plotting compass: the north-seeking point of the compass needle is attracted to the south pole and repelled by the north pole of the bar.

17.4 What rule determines the direction of force on a current-carrying conductor placed in a magnetic field? Explain briefly.

The direction of the force is given by Fleming's left-hand rule. Stretch the thumb, the forefinger and the middle finger of the left hand mutually at right angles. Point the forefinger along the direction of the magnetic field, and the middle finger along the direction of the current; then the thumb points in the direction of the force (the motion) on the conductor.

The force is given by $F = B\,I\,L \sin\theta$, where θ is the angle between the conductor and the field. It is greatest when the conductor is at right angles to the field ($\theta = 90°$, $F = BIL$) and zero when it lies along the field ($\theta = 0°$). This force is the basis of the electric motor, the loudspeaker and the moving-coil galvanometer.

17.5 What role does the commutator play in a D.C motor?

The commutator is a split ring fitted on the axle of the coil, with two carbon brushes pressing against it. It connects the rotating coil to the external circuit.

Its job is to reverse the direction of the current in the coil after every half rotation. Without it, the coil would turn only up to the vertical position and then stop (or turn back), because the force on each side would reverse. By changing over the connections twice in every cycle, the commutator keeps the force in the same rotational sense, so the coil goes on turning continuously in one direction. It thus acts as a mechanical rectifier for the motor.

17.6 What is the source of the Earth's magnetic field according to the theory?

According to the accepted dynamo theory, the Earth's magnetic field is produced by the motion of molten iron and nickel in its liquid outer core. This conducting fluid moves in convection currents and is also swirled by the Earth's rotation. A conductor moving in a magnetic field generates a current, and that current produces a magnetic field of its own — so the motion of the molten metal acts like a huge self-sustaining dynamo, with the solid inner core helping to shape the field.

The field behaves like that of a giant bar magnet tilted about 11° from the axis of rotation, with its magnetic south pole near the geographic north. The Earth's field is

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not constant: it drifts, and geological records show that it has reversed its direction many times in the past.

C. Constructed Response Questions

17.1 Why does a magnetic field form around a conductor when current flows through it, but not when the current is OFF? What does this reveal about the nature of electricity and magnetism?

When current flows, the charges in the conductor are moving. A moving charge is a tiny source of magnetic field — each electron in motion has its own magnetic field around it, and because all the electrons in the wire drift in the same direction, their small fields add up to give a net magnetic field that circles the wire. This was discovered by Oersted in 1820: a compass needle placed near a wire is deflected only when the current is switched on.

When the current is off, the electrons are not drifting in a common direction. Their magnetic effects cancel out, so no net field is present around the wire, and the compass needle stays still.

What this reveals: electricity and magnetism are not separate phenomena but two aspects of one electromagnetic interaction. A moving charge produces a magnetic field (current → magnetism), and a changing magnetic field produces a current (magnetism → electricity, electromagnetic induction). For this reason we speak of the electromagnetic field. The same idea explains why a current-carrying coil behaves like a magnet, why an electromagnet can be switched on and off, and why generators and transformers are possible.

17.2 In a D.C motor, why does the coil stop turning after 90° if the direction of the current is not reversed? What mechanical feature solves this issue?

In a motor the coil is placed between the poles of a magnet. While the plane of the coil is parallel to the field, the two sides of the coil carry current in opposite directions, so the forces on them are in opposite directions and the coil experiences a turning couple. The coil rotates.

After it has turned through 90° the plane of the coil becomes perpendicular to the field. In this position the two sides are moving parallel to the field, so the force on them ($F = BIL \sin\theta$, with $\theta = 0$) becomes zero and no turning effect is left. If the current direction is not changed, the coil would simply stop in this vertical position; if it overshoots a little, the force would reverse and push it back the other way, so it would oscillate and stop.

The mechanical feature that solves this is the split-ring commutator. It reverses the connections of the coil to the battery at the moment the coil passes the vertical position, so the current in each side of the coil is reversed. The forces therefore reverse at exactly that instant and keep pushing the coil round in the same direction, giving continuous rotation.

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17.3 How does placing a wire at different angles to a magnetic field affect the magnetic force it experiences, and why is this important in designing motors?

The force on a current-carrying wire in a magnetic field is F = B I L sin θ, where θ is the angle between the wire and the field. So the force depends strongly on the orientation of the wire.

When the wire is at right angles to the field (θ = 90°), sin θ = 1 and the force is maximum, F = B I L. When it is parallel to the field (θ = 0° or 180°), sin θ = 0 and the force is zero — the wire feels nothing. At an intermediate angle the force is proportional to sin θ, so it is smaller than the maximum.

Importance in motor design: the coil of a motor must be wound so that its sides cut the field lines as nearly at right angles as possible, because that is where the largest turning force is produced. But as the coil turns, the angle changes continuously, and at 90° of rotation the force falls to zero — this is exactly why a commutator is needed to keep the torque in the same direction. Designers also shape the pole pieces and place the coil in the strongest part of the field, and use several coils (armature windings) at different angles, so that the torque stays nearly constant and the motor runs smoothly instead of jerking.

D. Comprehensive Questions

17.1 How is a magnetic field created around a current-carrying wire, and what factors affect its strength?

Every electric charge in motion produces a magnetic field around itself. In a wire carrying current, billions of free electrons drift in the same direction, and their individual magnetic fields add up to give a net field whose lines are concentric circles around the wire. The field is thus created by the motion of charge, not by the charge itself — a stationary charge has only an electric field.

Factors affecting the strength of the field:

1. Current — the field is directly proportional to the current, B ∝ I; doubling the current doubles the field.

2. Distance from the wire — the field falls off inversely with the distance, B ∝ 1/r. It is strongest close to the wire.

3. Shape of the conductor — a straight wire gives a weak field spread over a large space, while a coil or solenoid concentrates the field lines inside it, so the field is much stronger. A toroid concentrates it even more.

4. Number of turns — for a coil the field is proportional to the number of turns per unit length, B = μ_0 n I, so more turns give a stronger field.

5. Core material — placing a soft-iron core inside the coil increases the field hundreds of times, because the iron becomes magnetised and adds its own field. This is how an electromagnet is made, and its strength can be controlled by changing the current.

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17.2 Explain the concept of the right-hand grip rule and how it helps determine the direction of the magnetic field around a current-carrying conductor.

The right-hand grip rule (Maxwell's corkscrew rule) is a rule for finding the direction of the magnetic field produced by a current. Imagine holding the conductor in the right hand with the thumb pointing along the direction of the conventional current; then the curled fingers of the hand point in the direction of the circular magnetic field lines around the wire.

How it is used: if the current in a vertical wire is flowing upwards, the thumb points up and the fingers curl anticlockwise when the wire is seen from above, so the field lines run anticlockwise. If the current is downwards, the field lines run clockwise. For a coil or solenoid the same rule applies in a modified form: grip the coil with the right hand so that the fingers curl in the direction of the current round the turns, and the thumb then points towards the north pole (the end from which the field lines come out).

The rule works because the magnetic field is always perpendicular to the current and to the radius from the wire. It is used to predict the field of any straight wire, loop or coil, to find which end of a solenoid becomes the north pole, and hence to design electromagnets, motors and relays correctly.

17.3 Describe an experiment that demonstrates the magnetic field produced by a current-carrying conductor using a compass or iron filings.

Apparatus: a long straight copper wire, a wooden board, a 6 V battery with a switch, a rheostat, some plotting compasses (or iron filings and a sheet of paper), and connecting leads.

Procedure with a compass. Pass the wire vertically through a hole in the board and connect it to the battery through the switch. Place a small plotting compass on the board close to the wire, with the switch open. Note the direction in which the compass needle points — along the Earth's field. Now close the switch and observe the needle: it turns at once and sets itself at right angles to the wire, along the tangent to the circle around the wire. Move the compass to different positions around the wire and mark the needle direction each time; the marks show that the field lines are concentric circles. Reverse the battery connections, and the needle turns through 180° — the field has reversed. Increase the current with the rheostat and the deflection becomes larger, showing a stronger field.

Procedure with iron filings. Pass the wire through the centre of a horizontal card and sprinkle iron filings thinly on the card. Switch on the current and tap the card gently. The filings arrange themselves in concentric circles around the hole, showing the shape of the field. On reversing the current the pattern of the circles stays the same (the direction reverses, which the compass shows).

Conclusion: a current-carrying conductor is surrounded by a magnetic field whose lines are circles centred on the wire; the direction of the field is given by the right-hand grip rule, and its strength increases with the current and decreases with distance from the wire.

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17.4 Discuss the theory behind the Earth's magnetic field and the role of the Earth's molten core in generating it.

The Earth behaves as if a huge bar magnet were placed inside it, with its magnetic poles close to the geographic poles. The field is not due to any permanent magnet in the rock, because the temperature deep inside the Earth is far above the Curie point, at which iron loses its magnetism.

Dynamo theory. The accepted explanation is the dynamo (geodynamo) theory. The outer core of the Earth, between about 2900 km and 5150 km depth, is liquid iron and nickel — a good conductor of electricity. Heat from the solid inner core and from radioactive decay sets up powerful convection currents in this fluid, and the Earth's rotation makes them spiral and form helical flows. A conductor moving through a magnetic field generates a current (electromagnetic induction); the induced currents in the molten metal produce their own magnetic field, and the process becomes self-sustaining. The solid inner core, rotating at a slightly different rate, helps to organise the flow and to keep the field aligned with the axis of rotation.

Evidence and consequences: the field is roughly dipolar, tilted about 11° to the axis, and its magnetic south pole lies near the geographic north. Studies of magnetised minerals in old rocks show that the field has reversed its polarity many times, at intervals of about 10^5–10^6 years, and it also drifts slowly westward. The magnetic field protects life by deflecting the charged particles of the solar wind, which is why the aurora is seen near the poles, and it is used by navigators with a compass.

Points to Remember

- A current always produces a magnetic field — right-hand grip rule gives its direction.
- B ∝ I and B ∝ 1/r for a straight wire; $B = \mu_0 nI$ inside a solenoid.
- Force on a conductor: F = BIL sin θ; direction by Fleming's left-hand rule.
- D.C. motor: electrical → mechanical; the commutator reverses the current every half-turn.
- Earth's field: dynamo action of molten iron in the outer core.